

What happened during the COVID-19 pandemic?

Returns, portfolio risk and dependence in ten US stocks

ME317: Statistical Methods for Risk Management

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Abstract

I study ten US stocks from five sectors using daily adjusted prices from 2018–2021. The 2018–2020 sample contains the main COVID-19 shock and is used to estimate the models; 2021 is kept aside for backtesting. Returns are strongly non-normal and large moves became much more synchronised in March 2020. For a fixed \$10,000 buy-and-hold portfolio, the normal model gives much smaller 99% tail losses than historical simulation or a Student- t model. A Student- t copula gives the best likelihood-based fit and, unlike a Gaussian copula, allows positive crash dependence. The exercise shows that conclusions about risk depend as much on the assumed tails and dependence as on ordinary volatility.

1 Data and construction

The sample contains AAPL and MSFT (technology), JPM and BAC (financials), XOM and CVX (energy), WMT, KO and DIS (consumer), and BA (industrials). I first downloaded daily observations from Yahoo Finance with the `getSymbols` function in `quantmod` on 22 July 2026 and rechecked them on 24 July; all reported results were unchanged at the displayed precision. The raw object has date, open, high, low, close, volume and adjusted close for each ticker; the analysis keeps one *adjusted close* P_{it} per stock. Yahoo defines it as the close corrected for applicable splits and dividend distributions [1]. This avoids treating a split or an ex-dividend adjustment as a genuine market loss and makes the price paths total-return-equivalent series. The code records the download date and prints `sessionInfo()` so that the R and package versions are available.

The merged data set runs from 2 January 2018 to 31 December 2021 and contains 1,008 common trading days. There were no duplicate dates and no incomplete merged rows. Weekends and exchange holidays are not missing observations, so I did not interpolate them. Had one ticker been absent on a day, the code would have removed that whole row with `na.omit`, keeping returns synchronous. I also inspected the largest absolute returns. They cluster around dated COVID-19 news and are economically credible, so none was winsorised or deleted.

Daily log-returns are

$$X_{it} = \log(P_{it}) - \log(P_{i,t-1}).$$

There are 755 in-sample return vectors (2018–2020) and 252 out-of-sample vectors (2021). The earlier years provide a calm comparison with the pandemic and prevent the model from being estimated on the crash weeks alone.

Table 1: Daily log-return risk and dependence across regimes. Volatility entries are sample SDs; portfolio returns use the fixed positions defined in Section 3.

Measure	2018–2019	Mar–Apr 2020	May–Dec 2020
Return observations (n)	502	43	170
Mean stock volatility (%)	1.42	6.15	2.39
Mean pairwise correlation	0.41	0.72	0.42
Portfolio return volatility (%)	0.99	4.98	1.50
Worst portfolio log-return (%)	4.64	13.25	7.01

The periods represent a pre-pandemic baseline, the acute March–April shock and the subsequent 2020 recovery. Table 1 makes the regime change quantitative. During the shock, mean stock volatility rose more than fourfold, correlation increased from 0.41 to 0.72 and portfolio return volatility rose from 0.99% to 4.98%. From May onward correlation returned close to its pre-pandemic value, but volatility remained elevated. Worst log-returns are descriptive, because the three windows contain different numbers of observations.

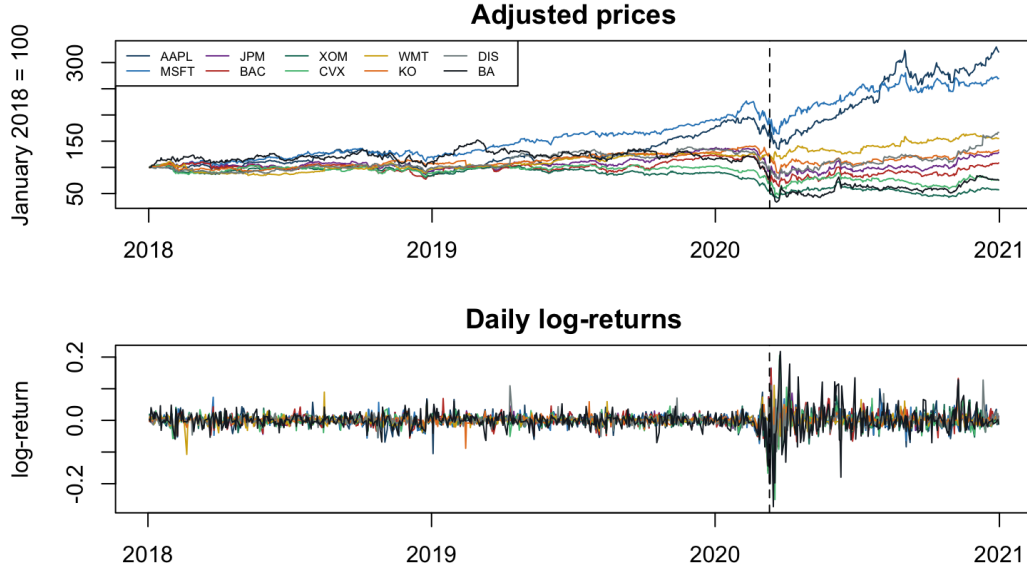


Figure 1: Adjusted prices rebased to 100 and daily log-returns. The dashed line is the WHO pandemic declaration on 11 March 2020.

2 Returns and extreme events

Figure 1 shows both the level effect and volatility clustering. Technology recovered quickly and ended 2020 well above its starting level, while energy and BA remained depressed. In returns, the main feature is not a permanent change in the mean but a dense block of unusually large positive and negative observations in March 2020.

The largest absolute in-sample moves were XOM -13.04% on 9 March; KO -10.17% and DIS -13.91% on 12 March; JPM $+16.56\%$ on 13 March; AAPL -13.77% , MSFT -15.95% , BAC -16.72% and BA -27.24% on 16 March; WMT $+11.07\%$ on 17 March; and CVX -25.01% on 18 March. Thus every stock’s maximum occurs within the 9–18 March market shock, although the signs differ as prices whipsawed.

The cross-stock average fell 10.62% on 9 March, 11.32% on 12 March and 13.90% on 16 March; it then rose 11.50% on 24 March. These dates are not arbitrary outliers. The 9 March fall coincided with the COVID demand shock and the suspension of OPEC+ production cuts; WTI fell 25% that day [2]. WHO characterised COVID-19 as a pandemic on 11 March [3], followed by the broad 12 March sell-off. On 15 March the Federal Reserve cut its target range to $0-0.25\%$ and announced large asset purchases [4]; the next session, 16 March, contains the largest move for AAPL, MSFT, BAC and BA. The 24 March rebound followed the Fed’s additional asset purchases and measures to support credit announced the previous day [5]. Finally, the 7.31% average rise on 9 November coincides with Pfizer–BioNTech’s first announcement of more than 90% vaccine efficacy [6]. These links are timing evidence, not a claim that each daily return has only one cause.

Table 2: In-sample daily return diagnostics. SD is in percent; normal kurtosis is 3.

Stock	SD	Skew.	Kurt.	JB p
AAPL	2.21	-0.39	8.90	$< 2.2 \times 10^{-16}$
MSFT	2.03	-0.39	12.45	$< 2.2 \times 10^{-16}$
JPM	2.24	-0.09	16.24	$< 2.2 \times 10^{-16}$
BAC	2.46	-0.06	14.25	$< 2.2 \times 10^{-16}$
XOM	2.19	-0.13	10.37	$< 2.2 \times 10^{-16}$
CVX	2.51	-1.20	28.82	$< 2.2 \times 10^{-16}$
WMT	1.52	0.45	16.58	$< 2.2 \times 10^{-16}$
KO	1.51	-0.95	12.03	$< 2.2 \times 10^{-16}$
DIS	2.09	0.48	14.45	$< 2.2 \times 10^{-16}$
BA	3.55	-0.48	16.57	$< 2.2 \times 10^{-16}$

Table 2 gives the quantitative reason not to use a Gaussian model without checking it. Every kurtosis estimate

is far above 3. CVX is the clearest left-skewed series, whereas WMT and DIS are mildly right-skewed, so the departure is not identical across sectors. Jarque–Bera rejects marginal normality for every stock. The CVX and BA QQ-plots in Figure 2 bend far away from the line in both tails; CVX has the stronger left tail. Thus rejection is driven mainly by fat tails, with skewness adding to it for some stocks.

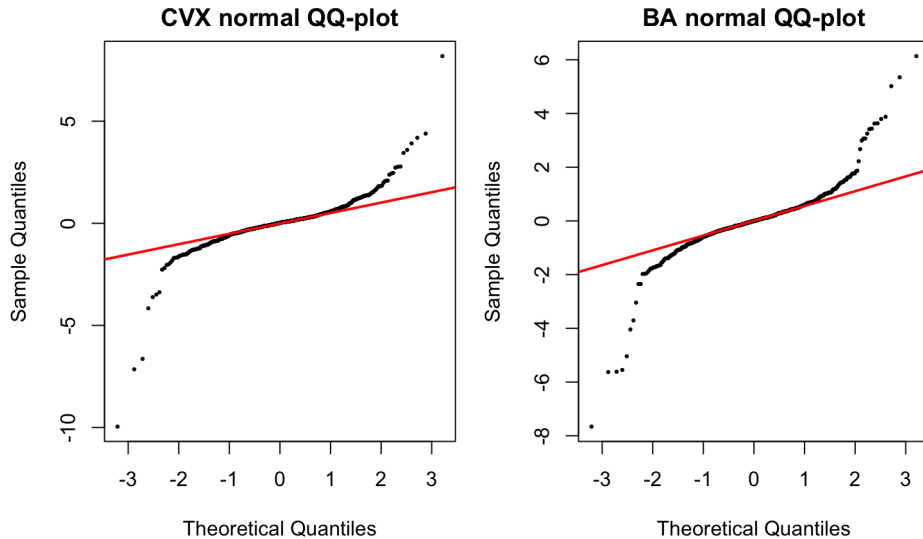


Figure 2: Normal QQ-plots for two of the most non-Gaussian returns.

The scatter plots in Figure 3 add an important dependence point. Same-sector pairs are tight: correlations are 0.77 for AAPL–MSFT, 0.94 for JPM–BAC, and 0.84 for XOM–CVX. MSFT–XOM is looser at 0.47, and the weakest pair among all 45 pairs is WMT–BA at 0.21. Sector diversification is therefore real in ordinary periods. However, the extra mass in the lower-left corners shows that diversification weakened on the worst days. A single correlation coefficient cannot describe this change in tail behaviour.

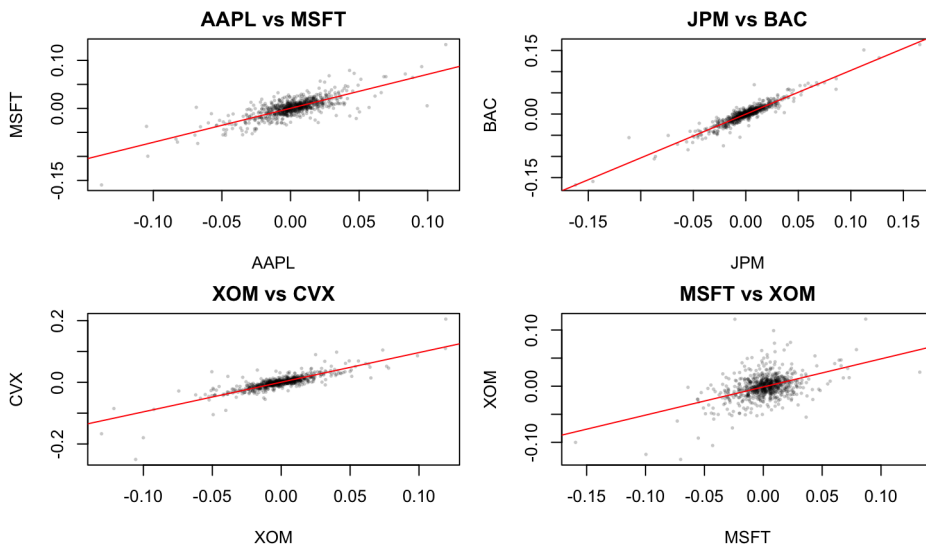


Figure 3: Representative within-sector and cross-sector return scatter plots.

3 Portfolio risk and backtesting

At the first date I invest \$1,000 in each series. If $h_i = 1000/P_{i0}$, the positions h_i are not rebalanced and the exact daily dollar loss is

$$L_t = -\sum_{i=1}^{10} h_i (P_{it} - P_{i,t-1}).$$

Here h_i denotes fixed units of the adjusted-price series, not a literal number of shares valued from the unadjusted close. The resulting calculation is therefore a total-return-equivalent buy-and-hold portfolio. For

each confidence level α , historical VaR is the sample α -quantile and historical ES is the mean loss above it. For the normal model I use $\hat{\mu} + \hat{\sigma}\Phi^{-1}(\alpha)$ and $\hat{\mu} + \hat{\sigma}\phi(\Phi^{-1}(\alpha))/(1-\alpha)$. The heavy-tailed model is $L = \mu + sT_\nu$, fitted by maximum likelihood; its estimated degrees of freedom are only $\hat{\nu} = 2.31$.

Table 3: One-day portfolio risk estimated on 2018–2020, in dollars.

Level	Historical		Normal		Student- <i>t</i>	
	VaR	ES	VaR	ES	VaR	ES
95%	251.8	462.1	283.6	357.3	222.9	433.8
99%	538.5	825.2	403.8	463.6	504.2	913.9

A historical 99% VaR of about \$539 means that, in the estimation sample, only 1% of daily losses were larger than this threshold. Conditional on crossing it, the average loss was much higher—about \$825—which is the historical ES. At 95%, the VaR ordering is not monotone: fitting a very heavy-tailed *t* distribution puts more probability both near the centre and far in the tails, so its 95% quantile is comparatively low. ES already tells a less reassuring story. At 99%, the normal model is clearly the least conservative: its ES is \$464, versus \$825 empirically and \$914 under Student-*t*. In other words, matching only a mean and variance misses the severity of the March observations. The large Student-*t* ES also shows estimation sensitivity: with ν close to 2, a few tail points have substantial influence.

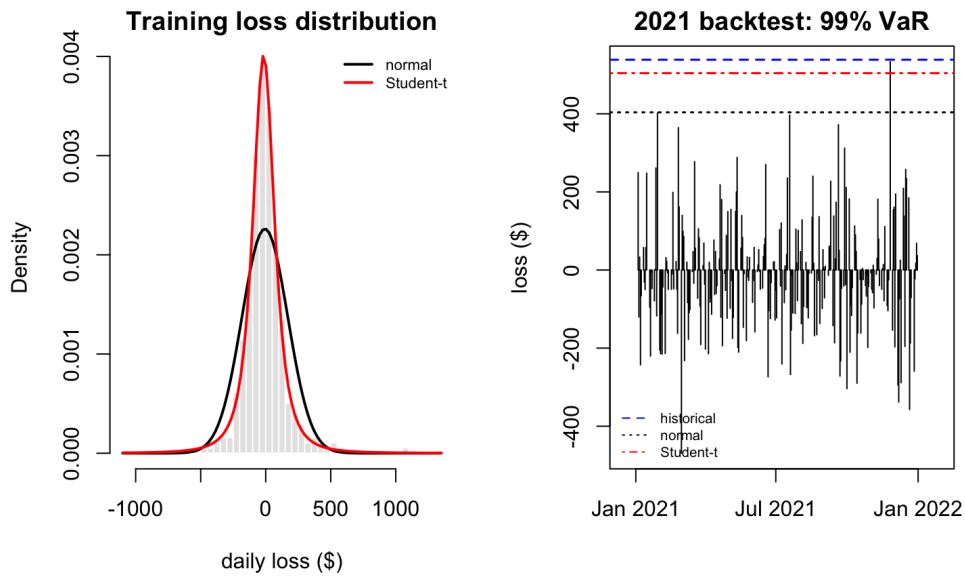


Figure 4: Fitted loss densities (left) and the 2021 99% VaR backtest (right).

For the out-of-sample test I keep both the positions and 2018–2020 estimates fixed. A violation is $I_t = 1\{L_t \geq \text{VaR}_\alpha\}$. I use Kupiec’s likelihood-ratio test of the null $P(I_t = 1) = 1 - \alpha$, with a 5% test level. At 95%, 12.6 exceptions are expected: historical, normal and Student-*t* produce 11 ($p = .637$), 7 ($p = .078$) and 17 ($p = .226$), respectively. None is rejected, although the normal model is close. At 99%, 2.52 are expected. Historical simulation has zero exceptions, so the asymptotic Kupiec test rejects it ($p = .024$); normal and Student-*t* each have one ($p = .273$) and pass. However, the expected count is small. As a sensitivity check, an exact two-sided binomial test for the historical model gives $p = .190$, and the probability of observing no exceptions under the null is 7.9%. The conclusion that historical VaR is excessively conservative is therefore sensitive to the small-sample approximation.

This does not make normality true. The historical estimate was dominated by the exceptional 2020 regime, while 2021 was calmer. Conversely, a single exception gives little power to distinguish the other models. Kupiec checks the total frequency but not clustering. The result is best read as evidence of *model risk*: the ranking of methods changes with the confidence level, evaluation year and small-sample treatment.

4 Copulas and crash dependence

A copula separates two questions: how heavy each stock’s individual tails are, and how likely several stocks are to fall together. To study the second question after removing the margins, I transform each return to

ranks $U_{it} = \text{rank}(X_{it})/(n + 1)$ and fit four 10-dimensional copulas. Gaussian, Clayton and Gumbel are estimated by maximum pseudo-likelihood. Student- t uses the lecture method `itau.mpl`: its 45 correlations are obtained from Kendall’s tau and only the degrees of freedom are then estimated by pseudo-likelihood. Hence its reported likelihood is evaluated at this hybrid estimate, not at a joint 46-parameter maximum. Gaussian and Student- t use unstructured correlation matrices; Clayton and Gumbel use one exchangeable dependence parameter.

Table 4: Copula fit. Larger log-likelihood and smaller AIC/BIC are better; values are rounded.

Model	Parameters	Log-likelihood	AIC	BIC
Gaussian	45	2461	-4832	-4624
Student- t	46	2774	-5455	-5242
Clayton	1	1352	-2703	-2698
Gumbel	1	1147	-2292	-2287

Since the models have very different numbers of parameters, the raw likelihood is not used alone. The AIC and BIC are calculated from each returned log-likelihood and its effective parameter count; for Student- t they are pragmatic pseudo-likelihood criteria rather than scores from a full joint maximum. Both criteria still favour Student- t : its AIC and BIC are about -5455 and -5242 , compared with -4832 and -4624 for Gaussian. Its estimated degrees of freedom are 4.05, indicating heavy joint tails. The exchangeable Clayton and Gumbel models cannot reproduce the large difference between, for example, bank–bank and consumer–industrial dependence. Gaussian is flexible in ordinary correlation but has zero asymptotic tail dependence.

For a t copula the lower- and upper-tail coefficients are equal. Across the 45 fitted pairs, the lower-tail coefficient ranges from 0.130 to 0.673, with median 0.214. It is highest for JPM–BAC and lowest for WMT–BA, which agrees with the scatter plots; the code now prints both pair labels with their coefficients. Therefore the selected copula assigns a positive limiting probability that one stock is crashing given that another is crashing. Clayton also has lower-tail dependence, but fits the full data much worse; Gumbel puts dependence in the upper return tail, and Gaussian has none. This matters directly for the portfolio: simultaneous falls remove the diversification assumed by a Gaussian model and push both VaR and, especially, ES upward.

I quantify this effect at 31 December 2020. For the Gaussian and Student- t copulas, I simulate 1,000,000 return vectors using the same stock-specific 2018–2020 empirical margins and the same fixed positions. With P_{iT} denoting the adjusted price at the evaluation date, simulated loss is

$$L = - \sum_{i=1}^{10} h_i P_{iT} \{\exp(R_i) - 1\}.$$

Table 5: Simulated one-day portfolio risk at 31 December 2020 under identical empirical margins, in dollars.

Dependence model	99% VaR	99% ES
Gaussian copula	604.8	783.7
Student- t copula	632.4	904.3

Changing the fitted dependence model raises simulated VaR from \$605 to \$632 and raises ES by about 15%, from \$784 to \$904. The larger ES under the Student- t copula is consistent with its positive joint tail dependence. These values should not be compared mechanically with Table 3: that table estimates the historical distribution of dollar losses over 2018–2020, whereas Table 5 values the positions held at the end of 2020.

5 Conclusion and limitations

The main empirical fact is a joint regime change, not ten isolated bad observations. During the acute shock, average pairwise correlation rose from 0.41 to 0.72; with identical empirical margins, the Student- t copula subsequently produced a 99% ES about 15% above the Gaussian-copula estimate. Pandemic news generated fat marginal tails and much stronger co-movement, which explains why a normal loss model looks reassuring precisely where risk management matters most. The Student- t model and copula are more consistent with these features, although their tail estimates are necessarily imprecise.

The analysis has four limits. Yahoo data are free and may be revised; the ten surviving firms also create survivorship bias. Static iid models ignore volatility clustering and do not update after 2020. The t copula forces symmetric upper and lower tail dependence. Because Clayton and Gumbel are exchangeable while Gaussian and Student- t use unstructured correlations, their comparison reflects both copula family and dependence-structure flexibility. Finally, one test year contains too few 99% violations for a powerful asymptotic comparison. A natural extension would use rolling volatility and a conditional-coverage test, but adding them here would change the scope and obscure the requested comparison of the three basic models.

References

- [1] Yahoo Finance Help, [What is the adjusted close?](#)
- [2] US Energy Information Administration, [Oil market volatility is at an all-time high](#), 27 March 2020.
- [3] World Health Organization, [Director-General's opening remarks](#), 11 March 2020.
- [4] Federal Reserve Board, [FOMC statement](#), 15 March 2020.
- [5] Federal Reserve Board, [Additional measures to support the economy](#), 23 March 2020.
- [6] Pfizer and BioNTech, [First interim Phase 3 analysis](#), 9 November 2020.

Appendix 1: Acknowledgement: The use of generative AI tools

I acknowledge the use of OpenAI Codex (GPT-5; OpenAI, <https://developers.openai.com/codex/>) to suggest including `sessionInfo()` to document the computational environment and support reproducibility, to locate and verify official sources, to provide feedback on the R code and the wording of the report, and to improve its LaTeX formatting. I made all final analytical, methodological and implementation decisions, adapted and implemented the final R code, and independently ran, reviewed and validated all data processing, calculations, simulations, tables, figures and conclusions. I remain fully responsible for the submitted work.

Appendix 2: Complete R code

```
# install.packages(c("quantmod", "moments", "tseries", "MASS", "copula"))
library(quantmod)
library(moments)
library(tseries)
library(MASS)
library(copula)
set.seed(202607)

# 1. Data
tickers <- c("AAPL", "MSFT", "JPM", "BAC", "XOM",
             "CVX", "WMT", "KO", "DIS", "BA")
sectors <- c("Technology", "Technology", "Financials", "Financials",
             "Energy", "Energy", "Consumer", "Consumer",
             "Consumer", "Industrials")
getSymbols(tickers, src = "yahoo",
           from = "2018-01-01", to = "2022-01-01")
# Data were first downloaded on 22 July and rechecked on 24 July 2026.
download.date <- as.Date("2026-07-22")
verification.date <- as.Date("2026-07-24")
print(c(first.download = download.date,
        verification = verification.date))
prices0 <- Ad(get(tickers[1]))
for(i in 2:length(tickers)) {
  prices0 <- merge(prices0, Ad(get(tickers[i])), all = TRUE)
}
colnames(prices0) <- tickers
# Keeping adjusted prices on common trading days.
cleaning <- c(rows.downloaded = nrow(prices0),
              rows_with_NA = sum(!complete.cases(prices0)),
              duplicate_dates = anyDuplicated(index(prices0)))
prices <- na.omit(prices0)
logprices <- log(prices)
returns <- na.omit(diff(logprices))
train <- returns["/2020-12-31"]
test <- returns["2021"]
X <- as.matrix(train)
print(cleaning)
print(c(first = as.character(first(index(prices))),
        last = as.character(last(index(prices))),
        rows = nrow(prices)))
# Jarque-Bera test for each column
JBTest <- function(dataMatrix)
{
  pvals <- rep(NA, ncol(dataMatrix))
  for(i in 1:ncol(dataMatrix))
  {
    pvals[i] <- tseries::jarque.bera.test(dataMatrix[, i])$p.value
  }
  return(pvals)
}
# Summary statistics.
stats <- data.frame(stock = tickers,
                    sector = sectors,
                    mean = rep(NA, length(tickers)),
                    sd = rep(NA, length(tickers)),
                    skewness = rep(NA, length(tickers)),
                    kurtosis = rep(NA, length(tickers)),
                    JB.p = JBTest(X))
for(i in 1:length(tickers)) {
  stats$mean[i] <- mean(X[, i])
  stats$sd[i] <- sd(X[, i])
  stats$skewness[i] <- moments::skewness(X[, i])
  stats$kurtosis[i] <- moments::kurtosis(X[, i])
}
rownames(stats) <- tickers
# Largest absolute return for each stock.
largest <- data.frame(stock = tickers,
                      date = as.Date(rep(NA, length(tickers))),
                      return = rep(NA, length(tickers)))
for(i in 1:length(tickers)) {
  j <- which.max(abs(X[, i]))
  largest$date[i] <- index(train)[j]
  largest$return[i] <- X[j, i]
}
print(round(stats[, 3:7], 4))
print(largest)
print(round(cor(X), 2))

# 2. Price, return, QQ and scatter plots.
dir.create("figures", showWarnings = FALSE)
cols <- c("#1b4f72", "#2e86c1", "#7d3c98", "#c0392b", "#117864",
          "#52be80", "#d4ac0d", "#e67e22", "#7f8c8d", "#17202a")
p.train <- prices["/2020-12-31"]
rebased <- sweep(coredata(p.train), 2,
                 as.numeric(p.train[, 1]), "/") * 100
par(mfrow = c(2, 1), mar = c(3, 4, 2, 1))
matplot(index(p.train), rebased, type = "l", lty = 1, col = cols,
        xlab = "", ylab = "January 2018 = 100",
        main = "Adjusted prices")
abline(v = as.Date("2020-03-11"), lty = 2)
legend("topleft", tickers, col = cols, lty = 1,
      cex = 0.55, ncol = 5)
matplot(index(train), X, type = "l", lty = 1, col = cols,
        xlab = "", ylab = "log-return",
        main = "Daily log-returns")
abline(v = as.Date("2020-03-11"), lty = 2)
par(mfrow = c(1, 2), mar = c(4, 4, 2, 1))
for(i in c(6, 10)) {
  qqnorm(scale(X[, i]),
        main = paste(tickers[i], "normal QQ-plot"),
        pch = 16, cex = 0.45)
  qqline(scale(X[, i]), col = "red", lwd = 2)
}
pairs.to.plot <- matrix(c("AAPL", "MSFT",
                          "JPM", "BAC",
                          "XOM", "CVX",
                          "MSFT", "XOM"),
                       ncol = 2, byrow = TRUE)
par(mfrow = c(2, 2), mar = c(4, 4, 2, 1))
for(i in 1:nrow(pairs.to.plot)) {
  s1 <- pairs.to.plot[i, 1]
  s2 <- pairs.to.plot[i, 2]
  plot(X[, s1], X[, s2], pch = 16, cex = 0.45,
       col = rgb(0, 0, 0, 0.22),
```



```

      xlab = s1, ylab = s2,
      main = paste(s1, "vs", s2))
abline(lm(X[, s2] ~ X[, s1]), col = "red")
}

# 3. Fixed buy-and-hold portfolio
# Invest $1,000 in each adjusted-price series at the first date.
# The fixed units therefore give a total-return-equivalent portfolio.
shares <- 1000 / as.numeric(prices[1, ])
portfolio.value <- as.numeric(coredata(prices) %*% shares)
portfolio.value <- xts(portfolio.value, order.by = index(prices))
portfolio.return <- na.omit(diff(log(portfolio.value)))
loss <- na.omit(-diff(portfolio.value))
loss.train <- as.numeric(loss["2020-12-31"])
loss.test <- as.numeric(loss["2021"])
# Compare the pre-pandemic, shock and recovery periods.
regime.names <- c("2018-2019", "Mar-Apr 2020", "May-Dec 2020")
regime.ranges <- c("2018-01-01/2019-12-31",
                  "2020-03-01/2020-04-30",
                  "2020-05-01/2020-12-31")
regime.table <- matrix(NA, nrow = 5, ncol = length(regime.ranges))
for(i in 1:length(regime.ranges)) {
  R <- returns[regime.ranges[i]]
  Rp <- as.numeric(portfolio.return[regime.ranges[i]])
  C <- cor(R)
  regime.table[i, ] <- c(nrow(R),
                        mean(apply(R, 2, sd)) * 100,
                        mean(C[lower.tri(C)]),
                        sd(Rp) * 100,
                        -min(Rp) * 100)
}
rownames(regime.table) <- c("Return observations (n)",
                          "Mean stock volatility (%)",
                          "Mean pairwise correlation",
                          "Portfolio return volatility (%)",
                          "Worst portfolio log-return (%)")
colnames(regime.table) <- regime.names
print(round(regime.table, 2))
# Fit a Student-t distribution by maximum likelihood.
t.fit <- fitdistr(loss.train, "t",
                 start = list(m = mean(loss.train),
                              s = sd(loss.train), df = 5),
                 lower = c(-Inf, 0.001, 2.01))
t.par <- t.fit$estimate
# Historical, normal and Student-t VaR and ES.
risk <- function(alpha)
{
  VaR.h <- as.numeric(quantile(loss.train, alpha))
  ES.h <- mean(loss.train[loss.train >= VaR.h])
  VaR.n <- mean(loss.train) + sd(loss.train) * qnorm(alpha)
  ES.n <- mean(loss.train) + sd(loss.train) *
    dnorm(qnorm(alpha)) / (1 - alpha)
  q <- qt(alpha, t.par["df"])
  VaR.t <- t.par["m"] + t.par["s"] * q
  ES.t <- t.par["m"] + t.par["s"] * dt(q, t.par["df"]) /
    (1 - alpha) * (t.par["df"] + q^2) / (t.par["df"] - 1)
  result <- c(Historical.VaR = VaR.h,
              Historical.ES = ES.h,
              Normal.VaR = VaR.n,
              Normal.ES = ES.n,
              Student.t.VaR = VaR.t,
              Student.t.ES = ES.t)
  return(result)
}
levels <- c(0.95, 0.99)
risk.table <- matrix(NA, nrow = length(levels), ncol = 6)
for(i in 1:length(levels)) {
  risk.table[i, ] <- risk(levels[i])
}
colnames(risk.table) <- c("Historical.VaR", "Historical.ES",
                        "Normal.VaR", "Normal.ES",
                        "Student.t.VaR", "Student.t.ES")
rownames(risk.table) <- paste0(100 * levels, "%")
print(round(risk.table, 1))
print(t.par)
# Kupiec proportion-of-failure test.
kupiec <- function(realised.loss, VaR, alpha)
{
  n <- length(realised.loss)
  hits <- sum(realised.loss >= VaR)
  phat <- hits / n
  LR <- 2 * (dbinom(hits, n, phat, log = TRUE) -
            dbinom(hits, n, 1 - alpha, log = TRUE))
  result <- c(exceptions = hits,
              rate = phat,
              LR = LR,
              p.value = 1 - pchisq(LR, 1))
  return(result)
}
model.names <- c("Historical", "Normal", "Student-t")
VaR.names <- c("Historical.VaR", "Normal.VaR", "Student.t.VaR")
backtest <- data.frame(alpha = rep(levels, each = 3),
                      model = rep(model.names, times = 2),
                      exceptions = rep(NA, 6),
                      rate = rep(NA, 6),
                      LR = rep(NA, 6),
                      p.value = rep(NA, 6))
k <- 1
for(i in 1:length(levels)) {
  for(j in 1:length(model.names)) {
    result <- kupiec(loss.test,
                    risk.table[i, VaR.names[j]],
                    levels[i])
    backtest[k, 3:6] <- result
    k <- k + 1
  }
}
backtest.print <- backtest
backtest.print[, c(1, 3, 4, 5, 6)] <-
  round(backtest.print[, c(1, 3, 4, 5, 6)], 4)
print(backtest.print)
# Exact-binomial sensitivity check for historical 99% VaR.
hist.hits.99 <- backtest$exceptions[
  backtest$alpha == 0.99 & backtest$model == "Historical"]
exact.99 <- binom.test(hist.hits.99, length(loss.test), p = 0.01,
                      alternative = "two.sided")
print(c(expected.exceptions = 0.01 * length(loss.test),
        exact.binomial.p = exact.99$p.value,
        probability.zero = dbinom(0, length(loss.test), 0.01)))
par(mfrow = c(1, 2), mar = c(4, 4, 2, 2))
hist(loss.train, breaks = 55, prob = TRUE,
     col = "grey90", border = "white",
     main = "Training loss distribution",
     xlab = "daily loss ($)")
curve(dnorm(x, mean(loss.train), sd(loss.train)),
      add = TRUE, lwd = 2)
curve(dt((x - t.par["m"]) / t.par["s"], t.par["df"]) /
      t.par["s"], add = TRUE, col = "red", lwd = 2)
legend("topright", c("normal", "Student-t"),
      col = c("black", "red"), lty = 1, lwd = 2,
      cex = 0.7, bty = "n")
backtest.dates <- index(loss["2021"])
plot(backtest.dates, loss.test, type = "h", xaxt = "n",
     xlim = as.Date(c("2020-12-15", "2022-01-31")),
     xlab = "", ylab = "loss ($)",
     main = "2021 backtest: 99% VaR")
axis.Date(1, at = as.Date(c("2021-01-01", "2021-07-01", "2022-01-01")),
          format = "%b %Y")
abline(h = risk.table["99%", VaR.names],
      col = c("blue", "black", "red"),
      lty = c(2, 3, 4), lwd = 1.5)
legend("bottomleft", c("historical", "normal", "Student-t"),
      col = c("blue", "black", "red"),
      lty = c(2, 3, 4), cex = 0.65, bty = "n")

# 4. Dependence and copulas
# Form the pseudo-sample using ranks
U <- matrix(NA, nrow = nrow(X), ncol = ncol(X))
for(i in 1:ncol(X)) {
  U[, i] <- rank(X[, i]) / (nrow(X) + 1)
}
colnames(U) <- tickers
d <- ncol(U)
# Gaussian copula.
fit.g0 <- fitCopula(normalCopula(dim = d, dispstr = "un"), U,
                   method = "itau", estimate.variance = FALSE)
fit.g <- fitCopula(normalCopula(dim = d, dispstr = "un"), U,
                  method = "mpl", start = coef(fit.g0),
                  estimate.variance = FALSE)
# Student-t, Clayton and Gumbel copulas.
# For Student-t, itau.mpl estimates correlations from Kendall's tau
# and then estimates the degrees of freedom by pseudo-likelihood.
fit.t <- fitCopula(tCopula(dim = d, dispstr = "un", df.fixed = FALSE), U,
                  method = "itau.mpl", estimate.variance = FALSE)
fit.c <- fitCopula(claytonCopula(dim = d), U,
                  method = "mpl", estimate.variance = FALSE)
fit.gu <- fitCopula(gumbelCopula(dim = d), U,
                  method = "mpl", estimate.variance = FALSE)
fits <- list(Gaussian = fit.g,
            Student.t = fit.t,
            Clayton = fit.c,
            Gumbel = fit.gu)
# Compare log-likelihood, AIC and BIC.
# For Student-t these are evaluated at the hybrid itau.mpl estimate.
copula.table <- data.frame(model = names(fits),
                          logLik = rep(NA, length(fits)),
                          parameters = rep(NA, length(fits)),
                          AIC = rep(NA, length(fits)),
                          BIC = rep(NA, length(fits)))
for(i in 1:length(fits)) {
  copula.table$logLik[i] <- as.numeric(logLik(fits[[i]]))
  copula.table$parameters[i] <- attr(logLik(fits[[i]]), "df")
  copula.table$AIC[i] <- -2 * copula.table$logLik[i] +
    2 * copula.table$parameters[i]
  copula.table$BIC[i] <- -2 * copula.table$logLik[i] +
    log(nrow(U)) * copula.table$parameters[i]
}
rownames(copula.table) <- copula.table$model
print(round(copula.table[, 2:5], 1))
# Student-t copula lower and upper tail dependence are equal.
nu <- as.numeric(tail(coef(fit.t), 1))
P <- getSigma(fit.t@copula)
lambda.matrix <- 2 * pt(-sqrt((nu + 1) * (1 - P) / (1 + P)),
                      df = nu + 1)
pair.index <- which(lower.tri(lambda.matrix), arr.ind = TRUE)
lambda <- lambda.matrix[lower.tri(lambda.matrix)]
print(c(df = nu,
        min.lambda = min(lambda),
        median.lambda = median(lambda),
        max.lambda = max(lambda)))
tail.pairs <- data.frame(
  pair = paste(tickers[pair.index[, 2]],
              tickers[pair.index[, 1]], sep = " - "),
  lambda = lambda)
extreme.pairs <- tail.pairs[
  c(which.min(tail.pairs$lambda), which.max(tail.pairs$lambda)), ]
rownames(extreme.pairs) <- c("minimum", "maximum")
print(extreme.pairs)
# Simulated end-2020 risk with identical empirical margins.
n.sim <- 1000000
position.T <- shares * as.numeric(tail(p.train, 1))
simulate.loss <- function(fitted.copula)
{
  U.sim <- rCopula(n.sim, fitted.copula@copula)
  R.sim <- matrix(NA, nrow = n.sim, ncol = ncol(X))
  for(i in 1:ncol(X)) {
    R.sim[, i] <- quantile(X[, i], U.sim[, i],
                        type = 1, names = FALSE)
  }
  L.sim <- -rowSums(sweep(exp(R.sim) - 1, 2, position.T, "*"))
  return(L.sim)
}
sim.risk <- function(L)
{
  VaR <- as.numeric(quantile(L, 0.99, type = 1))
  ES <- mean(L[L >= VaR])
  return(c(VaR = VaR, ES = ES))
}
set.seed(202607)
loss.g <- simulate.loss(fit.g)
loss.t <- simulate.loss(fit.t)
copula.risk <- rbind(Gaussian = sim.risk(loss.g),
                    Student.t = sim.risk(loss.t))
print(round(copula.risk, 1))
# Record R and package versions used for the analysis.
print(sessionInfo())

```